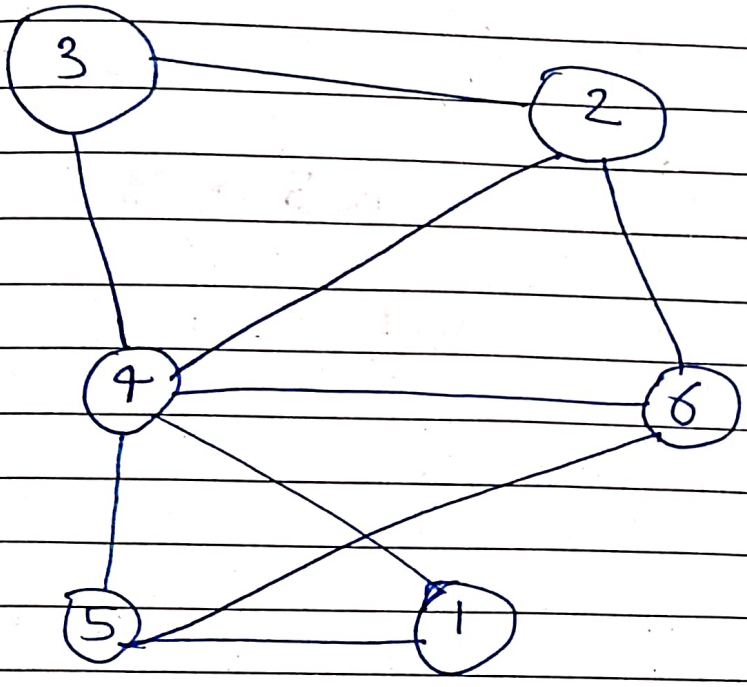
Network Science

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Tut-1



→ Degree

→ Closeness

→ Betweeness

→ Eccentricity

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→ Degree :

Node 1 : 2

Node 2 : 3

Node 3 : 2

Node 4 : 5

~~Node 5~~ :

Node 5 : 3

Node 6 : 3

→ Closeness (Formula : $\frac{n-1}{\sum d_{uv}}$)

~~$\sum d_{uv}$ =~~

Node 1 : $\sum d_{uv} = 1 + 1 + 2 + 2 + 2$

$\sum d_{uv} = 8$

$CC(1) = \frac{6-1}{8} = \frac{5}{8}$

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Node 2: $\sum d_{uv} = 1+1+1+2+2$

$\sum d_{uv} = 7$

$CC(2) = \frac{6-1}{7} = \frac{5}{7}$

Node 3: $\sum d_{uv} = 1+1+2+2+2$

$\sum d_{uv} = 8$

$CC(3) = \frac{5}{8}$

Node 4: $\sum d_{uv} = 1+1+1+1+1$

$\sum d_{uv} = 5$

$CC(4) = \frac{5}{5} = 1$

Node 5: $\sum d_{uv} = 1+1+1+2+2$

$\sum d_{uv} = 7$

$CC(5) = \frac{5}{7}$

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Node 6 : $\sum d_{uv} = 1+1+1+2+2$

$\sum d_{uv} = 7$

$CC(6) = \frac{5}{7}$

∴ Closeness → $CC(1) = \frac{5}{8}$, $CC(2) = \frac{5}{7}$, $CC(3) = \frac{5}{8}$

$CC(4) = 1$, $CC(5) = \frac{5}{7}$, $CC(6) = \frac{5}{7}$

→ Betweenness (Formula: $g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$

Node 1 ($v = 1$) : $g(1) = \sum_{s \neq t \neq v} \frac{\sigma_{st}(1)}{\sigma_{st}}$

→ Eccentricity:

Node 1 = max(1, 1, 2, 2, 2) = 2

Node 2 = max(1, 1, 1, 2, 2) = 2

Node 3 = max(1, 1, 2, 2, 2) = 2

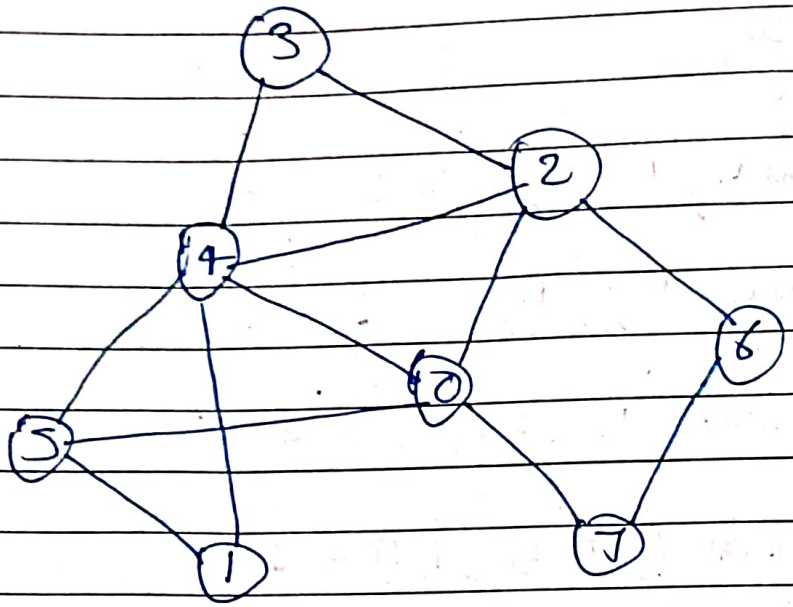
Node 4 = max(1, 1, 1, 1, 1) = 1

Node 5 = max(1, 1, 1, 2, 2) = 2

Node 6 = max(1, 1, 1, 2, 2) = 2

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Degree:

Node 0 → 4
Node 1 → 2
Node 2 → 4
Node 3 → 2
Node 4 → 5
Node 5 → 3
Node 6 → 2

Eccentricity:

Node 0 → max(1,1,1,1,2,2,2) = 2
Node 1 → max(1,1,2,2,2,3,3) = 3
Node 2 → max(1,1,1,1,2,2,2) = 2
Node 3 → max(1,1,2,2,2,2,3) = 3
Node 4 → max(1,1,2,1,1,2,2) = 2

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Node 5 → max(1, 1, 1, 2, 2, 2, 3) = 3
Node 6 → max(1, 1, 2, 2, 2, 3, 3) = 3
Node 7 → max(1, 1, 2, 2, 2, 3, 3) = 3

Closeness : $\left(\text{Formula} = \frac{n-1}{\sum d_{uv}}\right)$

~~Node (1)~~ : $CC(0) = \frac{7}{1+1+1+1+2+2+2} = \frac{7}{10}$

$CC(1) = \frac{7}{1+1+2+2+2+3+3} = \frac{7}{14} = \frac{1}{2}$

$CC(2) = \frac{7}{1+1+1+1+2+2+2} = \frac{7}{10}$

$CC(3) = \frac{7}{1+1+2+2+2+2+3} = \frac{7}{13}$

$CC(4) = \frac{7}{1+1+1+1+1+2+2} = \frac{7}{9}$

$CC(5) = \frac{7}{1+1+1+2+2+2+3} = \frac{7}{12}$

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$$CC(6) = \frac{7}{1+1+2+2+2+3+3} = \frac{7}{14} = \frac{1}{2}$$

$$CC(7) = \frac{7}{1+1+2+2+3+3} = \frac{7}{14} = \frac{1}{2}$$

